

Automated Food Calorie Estimation Using Image-Based Volume Analysis

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January 11, 2026

Problem Description

- Accurate calorie estimation is critical for dietary monitoring and health management
- Manual calorie logging is time-consuming and error-prone
- Image-based approaches offer a scalable alternative
- Core challenge: **estimating food volume from images**

Background Study

- Image-based dietary assessment relies on:
 - Food detection and classification
 - Portion size estimation
- Volume estimation is more challenging than recognition
- Most systems rely on geometric assumptions or multiple views

Existing Solution Categories

- Single-image (monocular) methods
- Two-image methods (top + side view)
- Multi-view methods (multiple side views)
- Depth sensor-based approaches (LiDAR / RGB-D)

Comparison of Existing Solutions

Method	Accuracy	User Effort	Robustness
Single Image	Low	Very Low	Low
Two Views	Medium	Medium	Medium
Multi-View	High	High	High
Depth Sensors	Very High	Medium	Very High

Motivation for Proposed Solutions

- No single existing method satisfies all constraints
- Need to explore trade-offs between:
 - Accuracy
 - User effort
 - Practical deployability
- Three candidate solution approaches are proposed

Proposed Solution Overview

- Common pipeline:
 - Food detection and segmentation
 - Surface area estimation
 - Height estimation
 - Volume and calorie computation
- Three alternative image acquisition strategies

Proposed Approach A: Single 45° Image

Input: One oblique image with thumb calibration

- YOLOv8-seg for food segmentation and classification
- Thumb used for scale calibration
- Table plane estimated using edge detection
- Area estimated via perspective rectification
- Height estimated from visible vertical extent

Approach A: Volume Computation

$$\text{Volume} = \text{Area} \times \text{Height}$$

- Area and height inferred from same image
- Fast but ill-conditioned estimation
- High variance in real-world usage

Proposed Approach B: Top View + One Side View

Input: Two images, thumb in both

- Top view:
 - YOLOv8-seg
 - Accurate surface area estimation
- Side view:
 - Plate baseline detection
 - Food silhouette extraction
 - Height estimation

Approach B: Key Advantages

- Decouples area and height estimation
- Lower error amplification
- Balanced accuracy and user effort
- Strong geometric justification

Proposed Approach C: Top View + Four Side Views

Input: One top view + four side views

- Thumb only in top view
- Scale transferred using plate geometry
- Height extracted from multiple viewpoints
- Aggregation using maximum or median height

Approach C: Key Advantages

- Most robust height estimation
- Handles occlusion and irregular shapes
- Lower variance and tighter error bounds
- Highest expected accuracy

Comparison of Proposed Approaches

Approach	Accuracy	User Effort	Complexity
A: Single Image	Low	Very Low	Low
B: Two Views	Medium	Medium	Medium
C: Multi-View	High	High	High

Economic Impact Analysis

- Uses commodity smartphone cameras
- No specialized hardware required
- Scalable for large user bases
- Potential integration with health apps

Environmental Impact Analysis

- Promotes food awareness and waste reduction
- Low additional energy consumption
- No physical measurement tools required

Social and Ethical Considerations

- Improved dietary awareness and health outcomes
- Privacy concerns with food images
- Bias in food recognition models
- Need for transparent error communication

Conclusion and Next Steps

- Food calorie estimation is a challenging but impactful problem
- Three candidate solutions explore accuracy–effort trade-offs
- Supervisor feedback will guide final approach selection
- Future work includes dataset collection and evaluation